Characterizing the Robustness of Musical Instrument Classification Under Acoustic Perturbations

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Introduction

Background

- Instruments' timbral properties are central to how music is composed, performed, and perceived. As a result, instrument classification is an active research area (musicology, style analysis, audio editing and retrieval) [1, 2].
- Since 2015, convolutional neural networks have pushed clean-audio classification accuracy from ~90% to as high as ~99% [7].
- **The Gap:** model performance is usually measured on clean, isolated audio. It remains unclear which models fail when real-world noise is introduced, under what noise, and which instruments break down first.

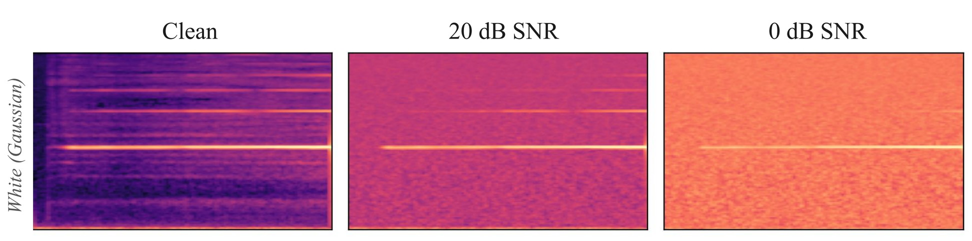

Fig.1 Flute A6 mel spectrogram; clean and under Gaussian noise

Objectives

- We run a controlled, head-to-head robustness comparison of six model families: handcrafted SVM; custom CNN and CRNN; and fine-tuned, pretrained MERT, AST, and PANNs.
- Trained on the same 12-instrument dataset, we hold the split and noise corpus identical across all models and add white (synthetic), audience (human sounds — ESC-50), and studio (real-world ambience from all 18 DEMAND environments) noise across an 8-level SNR sweep, measuring the change in classification performance.
- **Research Question:** How do instrument classification systems differ in their ability to retain classification performance under white, audience, and studio additive noise across a range of signal-to-noise ratios?

Methods

Source Recordings — Philharmonia Orchestra · 12 instruments · One labeled note / file
→ **Preprocessing** — Filter articulation · tile silence · Resample · one 3 s window · Normalize loudness · → 8,374 retained recordings
→ **Grouped Split** — Stratified by instrument, split by pitch 70/15/15 · Train 5,861 · Val 1,258 · Test 1,255
→ **Train Models** — Trained: SVM · CNN · CRNN · Fine-tuned: MERT · AST · PANNs
→ **Construct Noisy Test Set** — White (Gaussian), Audience (ESC-50), Studio (DEMAND) · 8 SNRs × 2 independent draws
→ **Evaluate / Score Models** — Identical noisy mixtures per model · Macro-F1 retention across SNR

Fig.2 Methods pipeline

- 6 models were trained; 3 handcrafted, trained-from-scratch; 3 pretrained approaches
- Validation data used to select stopping decisions and hyperparameters, but not final model fitting
- Test data held out during model training; each final model tested once on clean test set

Dataset & split

12	8,374	3 s
instruments	windows	@ 22.05 kHz
5,861	**1,258**	**1,255**
train	validation	test

Split is grouped by (instrument, note): every recording of a pitch lands in exactly one split, so near-duplicate takes cannot leak between train and test. A no-leak assertion runs on every dataset build.

Trained from scratch

No external pretraining

Model	Input to classifier	What's learned on this dataset
SVM	88 handcrafted audio features Timbre, spectrum, pitch, loudness	RBF decision boundary
CNN	128-band log-Mel spectrogram	Convolutional features + classifier
CRNN	128-band log-Mel spectrogram	Convolutional and recurrent Features plus classifier

Pretrained backbones

Pretraining corpus listed beneath each model

Model	Input to classifier	What's learned on this dataset
MERT Music audio	24 kHz waveform 13 time-averaged layer outputs	Layer weights + linear head Backbone stays frozen
AST AudioSet	16 kHz waveform Official extractor to log-Mel	Transformer backbone Plus 12-class head
PANNs AudioSet	32 kHz waveform CNN14 computes 64-band log-Mel	CNN14 backbone Plus 12-class head

Noise Construction

- Noise added to test set (1255 samples)
- SNR (Signal-to-noise ratio) is the ratio between the signal (audio source) compared to the noise. High SNR → faint noise; 0 dB SNR → noise is as loud as instrument

Eq. 1 · Power

$$P(x)=\frac{1}{N}\sum_{i=1}^{N}x_i^2$$

mean squared amplitude of a waveform

Eq. 2 · Seed

$$\text{seed}=\text{SHA256}(f|w|t|r)[:4]$$

fixes the noise draw (fingerprint, window, type, replicate)

Eq. 3 · Centering

$$n_c=n-\bar{n}$$

subtracts the mean to center the noise

Eq. 4 · Gain

$$\alpha=\sqrt{\frac{P(x)}{P(n_c)\cdot 10^{s/10}}}$$

multiplier that scales noise to the target SNR

Eq. 5 · Mixture

$$y=x+\alpha n_c$$

Eq. 6 · Check

$$\hat{s}=10\log_{10}\frac{P(x)}{P(y-x)}$$

- **white** — generated Gaussian noise — synthetic broadband control
- **audience** — ESC-50 human non-speech (targets 20–29): 400 clips, 10 classes — clapping, coughing, laughing, footsteps …
- **studio** — DEMAND: 18 real environments (domestic, nature, office, public, street, transport), one microphone per array

Every mixture is seeded by sha256(dataset fingerprint | window | noise type | replicate), so the 60,240-file corpus is bit-reproducible and identical for every model.

Results

Model	Clean macro-F1	AUC white	AUC audience	AUC studio
SVM	0.9788	0.2531	0.6352	0.5654
CNN	0.9708	0.3462	0.6336	0.4901
CRNN	0.9738	0.3215	0.6377	0.5017
MERT	0.8931	0.4321	0.6273	0.5773
PANNs	0.9885	0.3996	0.7689	0.7174
AST	**0.9908**	**0.6298**	**0.7834**	**0.7457**

Figure 5. Clean baseline and robustness AUC — area under the macro-F1-vs-SNR curve, dB-weighted (1.0 = perfect classification at every SNR). Bold marks the best value per column; model colours match Figure 6.

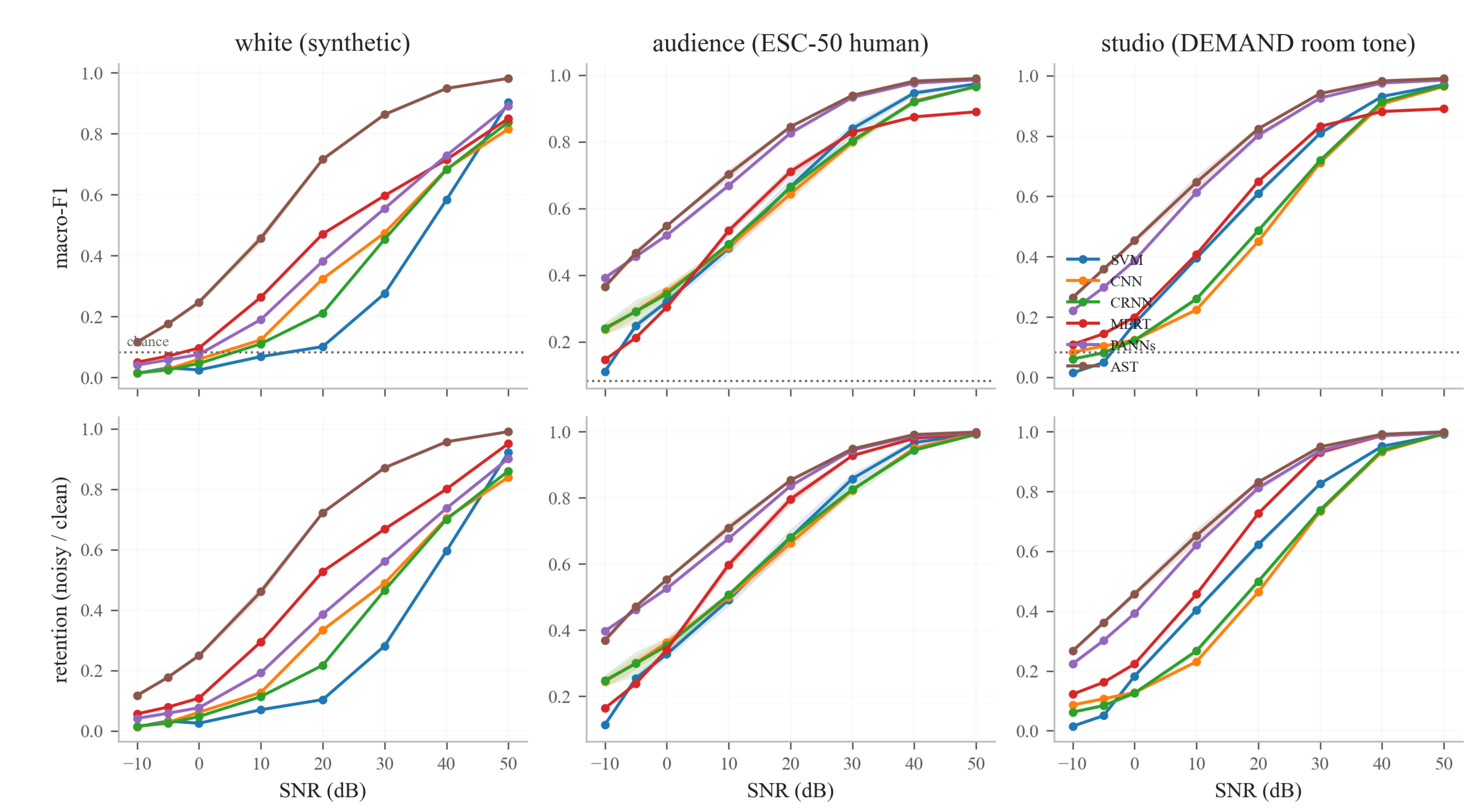

Figure 6. Robustness curves for all 6 models across 3 noise types (columns): macro-F1 (top row) and retention relative to clean score (bottom row), versus SNR from 50 to −10 dB. Shaded band is the spread across two noise draws; dotted line is 12-class chance (0.083).

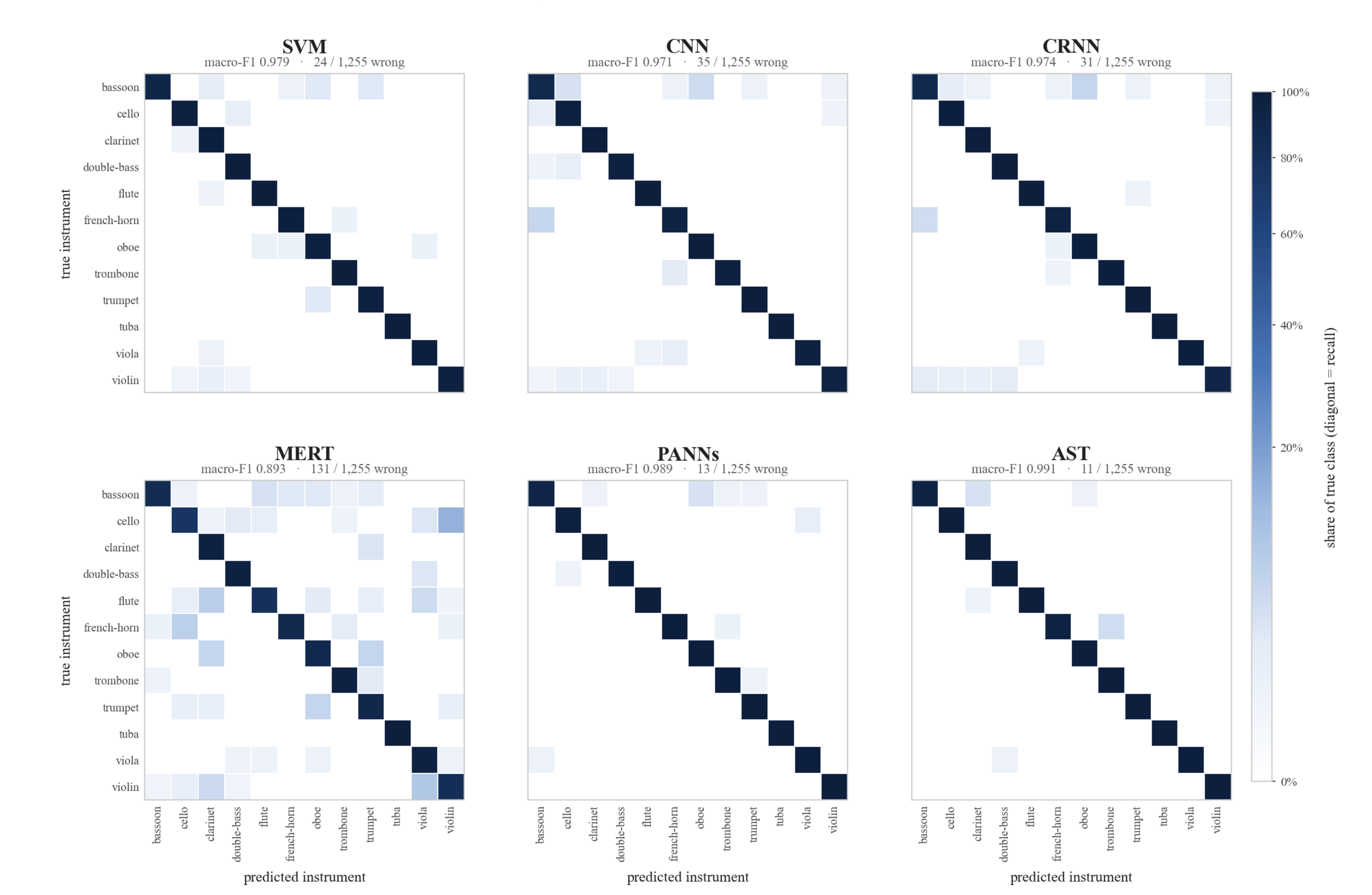

Figure 7. Clean-audio confusion for all six models, row-normalised: the diagonal is per-class recall, off-diagonal cells are the share of a true instrument sent elsewhere, on one colour scale across panels. Panel titles give clean macro-F1 and total misclassified windows.

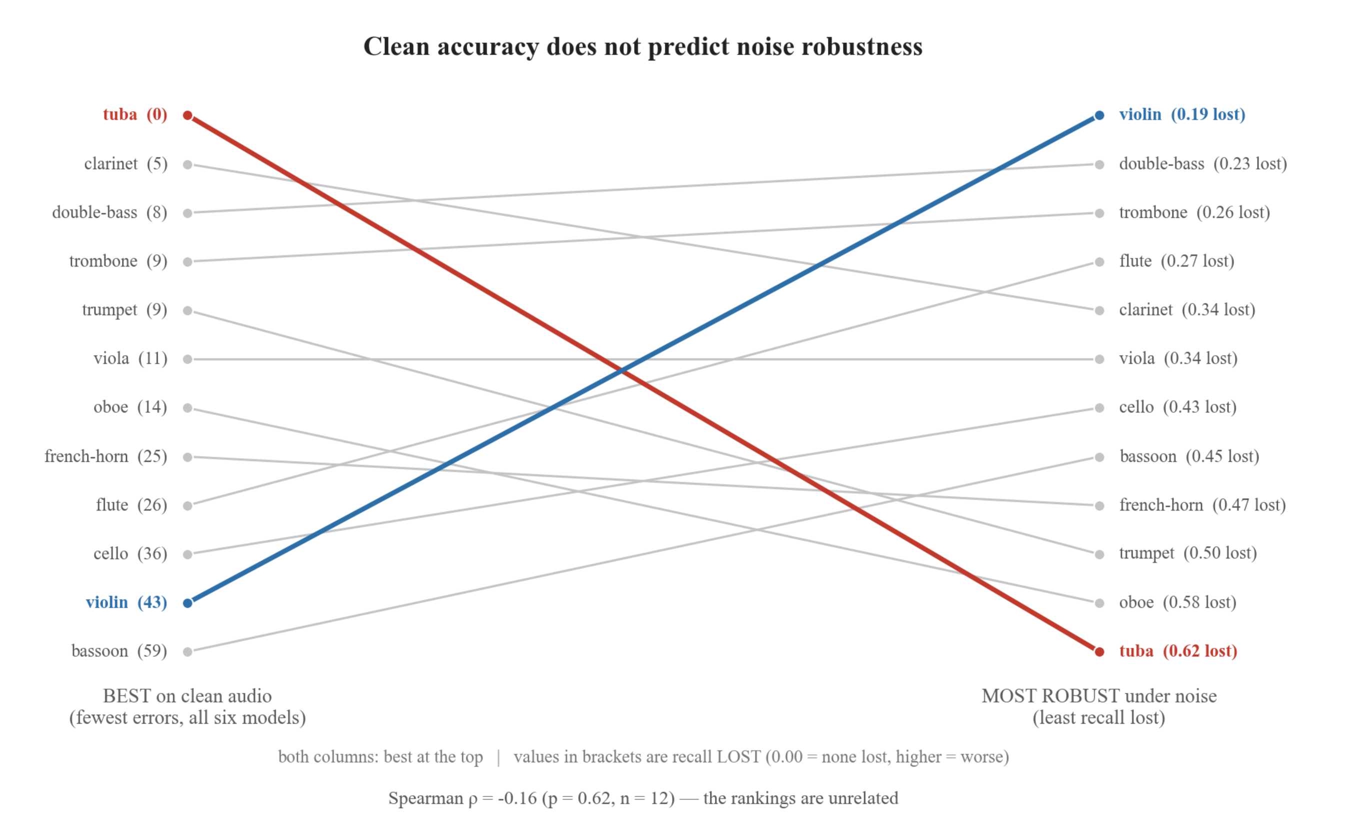

Figure 8. Instruments ranked by clean accuracy (left) and noise robustness (right), best at top. The rankings are unrelated (Spearman ρ = −0.16, p = 0.62). Tuba is classified perfectly by all six models but loses the most recall under noise (0.62); violin is 43 clean errors yet the most robust (0.19 lost).

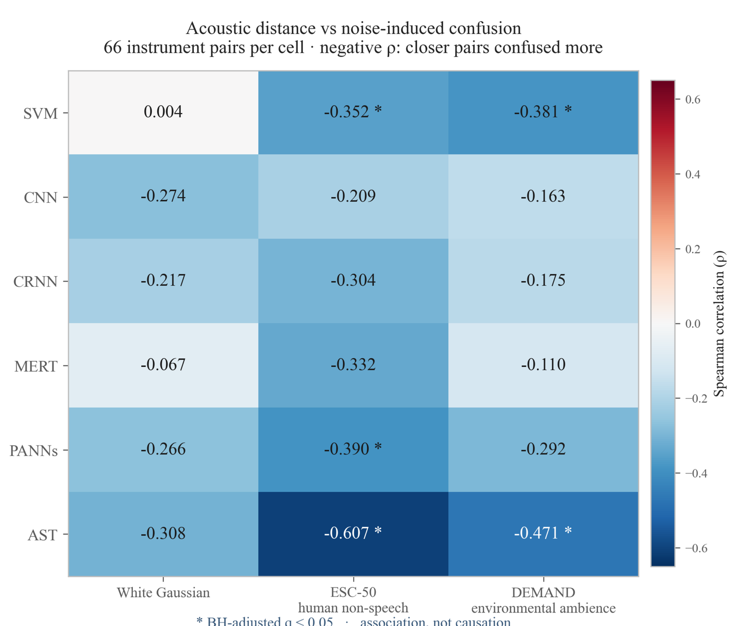

Figure 9a. Acoustic distance vs confusion associations across model and noise types

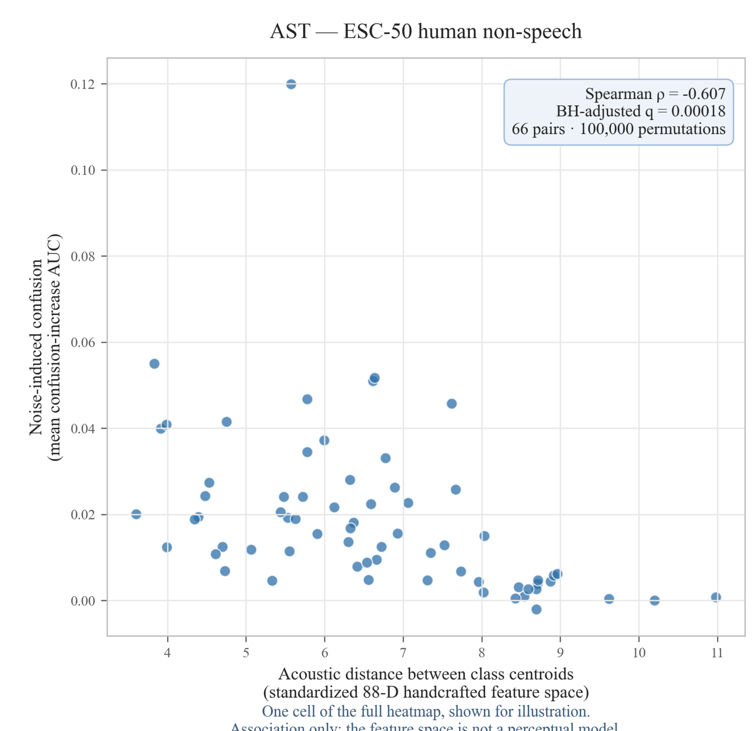

Figure 9b. Illustrative AST × ESC-50 human non-speech relationship across 66 instrument pairs

10% vs 68%
SVM keeps 10% of its clean score under white noise at 20 dB, but 68% under audience noise at the same loudness

0 errors, worst loss
tuba: perfect on clean audio for all six models, largest recall loss under every noise type

5 / 18
distance–confusion tests significant after Benjamini–Hochberg

Discussion

Evaluation Metrics

$$\text{Macro F1}=\frac{1}{12}\sum_{c=1}^{12}\text{F1}_c$$

Eq.7 Macro-F1 — primary evaluation metric; 12-instrument average

$$\text{Retention}=\frac{\text{Macro F1}_{noisy}}{\text{Macro F1}_{clean}}$$

Eq.8 Retention — standardized measure of robustness; how much clean score was retained after noise added

$$\text{AUC}=\frac{1}{s_{max}-s_{min}}\int_{s_{min}}^{s_{max}}\text{Retention}(s)\,ds$$

Eq.9 Area Under Robustness Curve — dB-weighted average macro-F1 under all eight noise levels

$$\text{Accuracy}=\frac{\#\,\text{correct}}{\#\,\text{examples}}$$

Eq.10 Accuracy — plain model correctness rate

Statistical Analysis

- Model differences were evaluated on identical test recordings; same clip with the same noise realizations
- Confidence intervals were estimated by resampling complete instrument-pitch groups
- Acoustic distance and noise-induced confusion were related using Spearman correlation. Significance assessed using 100,000 random instrument-label permutation tests, corrected using Benjamini–Hochberg correction to account for pure chance
- Paired comparisons used identical test windows and noise realizations. CNN exceeded CRNN by an average 0.1126 Macro-F1 over both realizations (0 & 1: 0.1122, 0.1130) under white noise at 20 dB SNR; 95% pitch-group bootstrap intervals r_0 = [0.0560, 0.1625] and r_1 = [0.0589, 0.1601] excluded zero
- Acoustically closer instrument pairs showed greater confusion; 5 of 18 model-noise tests are significant after Benjamini–Hochberg; strongest for AST under human non-speech (Spearman $\rho = -0.607$, permutation $p = 0.000010$, adjusted $q = 0.000180 < 0.05$).

Key Takeaways

- **Clean Macro-F1 ≠ robustness.** 5/6 models sat within 2 points on clean audio, but under white noise SVM fell from 3rd to last (AUC 0.25) and MERT rose from last to 2nd (0.43).
- **AST was the most robust model** under all 3 noise types (AUC 0.63 / 0.78 / 0.75), while the weakest model under each varied: SVM under white, MERT under audience, CNN under studio.
- **Pretrained Model ≠ robustness.** AST led each noise type, but MERT swung from 2nd under white to last under audience, and PANNs fell to 3rd under white.
- **Noise character:** audience and studio far less damaging than white. At 20 dB, retention under white vs audience: AST 72% vs 85%, SVM 10% vs 68%.
- **Instruments:** failures were uneven on both clean and noisy sweeps; tuba degraded most; acoustic distance analysis supports the “similar instruments confuse more” association, which is only correlational

Conclusion

- **Universal benchmark** for instrument classification models, tested on six frontier architectures
- **Evaluating robustness necessary**; clean accuracy alone is insufficient
- Noise type equally important as noise strength (SNR ratio)
- **Failures are structured.** Degradation concentrates in tuba, oboe, and trumpet. Confusion is predicted by acoustic distance in 5 of 18 model/noise tests, most strongly for AST.
- **Shared, seed-reproducible noise** makes model comparison possible.

Limitations

- Only one dataset (Philharmonia), restricting timbre to one instrument, player, and/or recording setup
- Recordings are isolated notes, not polyphonic music where several instruments play at once
- Most audio clips were tiled to fill the 3-second standard, which confounds content with repetition
- Nominal SNR and different frequencies of noise may cause instruments to be masked unequally
- SVM, AST, MERT and PANNs are single-seed runs, while CNN and CRNN are multi-seed spread, so small differences are not treated as effects

Future Work

- Polyphonic and multi-dataset audio
- Independently recorded real-world noise, more real acoustic issues
- More noise realizations to increase robustness against random draw
- Noise-aware training to further noise robustness

Reproducibility

- All per-window predictions, per-condition metrics, and the failure analysis are committed to the repository
- Every figure and table on this poster regenerates from one script each (scripts/*.py) against those files
- The noise corpus rebuilds bit-identically from the sealed dataset fingerprint — scan the GitHub QR below

References

Works Cited

GitHub

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